IC-1031 — Answer key and teaching guide (instructor only; do not share)

Data: synthetic, fixed seed 1031. The sheet prints each team's summary (Exhibit 1). mochi-latte-samples.csv holds the raw scores (12 teams × 3 recipes × 10 scores = 360 rows) for students to check after class. Every number below was computed by script and cross-checked against scipy (two-sample t, one-way ANOVA).

The truth (reveal only at the end)

Scores are normal with σ = 1.0. μ_A = 7.4, μ_B = 7.4, μ_C = 8.5.

- **Step 1: recipe A is above 7 (7.4 > 7), so H_a is true.**
  - **Teams that do not reject make a Type II error.**
  - **With n = 10 the power is only about 0.3, so most teams miss it. That is the lesson: small samples miss real but small effects.**
- **Step 2: A and B are the same, so H_0 is true.**
  - **A team that rejects makes a Type I error.**
  - **With α = 0.05 and 12 teams we expect about 0.6 such teams; here it is one (Team 05).**
- **Step 3: the recipes do differ (C is higher).**
  - **Teams that do not reject make a Type II error.**

Expected show of hands: Step 1, **4 reject, 8 do not**. Step 2, **1 rejects, 11 do not**. Step 3, **10 reject, 2 do not**.

How to run the 50 minutes

Changed 2026-10-04 after IC-1003, where most teams were stuck at the AI step (Step 2) and never reached the end. Three fixes:

1. **No AI and no file in class.** The summary numbers are printed on the sheet (Exhibit 1). Nobody can be stuck on loading data. The CSV is for checking after class.
2. **A time limit on every part, and a checkpoint after every step.** Use a visible timer. At "time", show the method on the board with one team's numbers, then move on. A stuck team copies the method and rejoins; nobody is left behind.
3. **The last step is a discussion, not more computing.** 15 minutes are protected for it.

Min	What	Your move at the checkpoint
0–3	Read the case aloud. Teams circle their row.	
3–13	Step 1, timer on each part	Cold-call one team for (b)–(d); write the method on the board. Show of hands. Do not explain the disagreement yet.
13–23	Step 2	Cold-call another team. Ask Team 05 (the only one that rejects) to read its t.

Min	What	Your move at the checkpoint
23–35	Step 3	Draw the empty ANOVA table on the board; one team fills it in aloud. Stress (b): MSE is the pooled variance again.
35–50	Step 4 discussion	2 minutes silent writing, then (a)–(c) with cold calls from both sides of the show of hands. Reveal. Last 3 minutes: (e), handed in. This is the graded part.

4.

If a step runs over, cut its last part, not the discussion. Walk the room during each part; look at what they write, and pick the next cold call from what you see.

Step 1: recipe A, one-sample t (df = 9)

(a) n = 10 is small and σ is unknown, so we estimate σ by s; the statistic follows a t distribution with n − 1 = 9 df (population assumed normal). (b) H_0: μ_A = 7, H_a: μ_A > 7. (d) Reject if t > 1.833.

Team	x̄_A	s_A	SE	t	p-value	Decision	vs truth	95% CI (extra)
Team 01	7.42	1.012	0.320	1.31	0.111	do not reject	**Type II error**	(6.70, 8.14)
Team 02	7.29	1.276	0.403	0.72	0.245	do not reject	**Type II error**	(6.38, 8.20)
Team 03	7.02	0.770	0.243	0.08	0.468	do not reject	**Type II error**	(6.47, 7.57)
Team 04	7.78	0.924	0.292	2.67	0.013	reject H0	correct	(7.12, 8.44)
Team 05	6.89	1.043	0.330	-0.33	0.627	do not reject	**Type II error**	(6.14, 7.64)
Team 06	7.21	0.837	0.265	0.79	0.224	do not reject	**Type II error**	(6.61, 7.81)
Team 07	7.94	0.574	0.181	5.18	0.000	reject H0	correct	(7.53, 8.35)

Team	x̄_A	s_A	SE	t	p-value	Decision	vs truth	95% CI (extra)
Team 08	6.90	0.537	0.170	-0.59	0.715	do not reject	**Type II error**	(6.52, 7.28)
Team 09	7.27	1.143	0.361	0.75	0.237	do not reject	**Type II error**	(6.45, 8.09)
Team 10	7.23	1.352	0.428	0.54	0.302	do not reject	**Type II error**	(6.26, 8.20)
Team 11	7.47	0.556	0.176	2.67	0.013	reject H0	correct	(7.07, 7.87)
Team 12	7.71	1.008	0.319	2.23	0.026	reject H0	correct	(6.99, 8.43)

Step 2: A against B, pooled two-sample t (df = 18)

(a) H_0: μ_A = μ_B (μ_A − μ_B = 0), H_a: μ_A ≠ μ_B. (b) df = 10 + 10 − 2 = 18. (d) Reject if |t| > 2.101. Assumptions (ask aloud): both populations normal, equal variances, independent random samples.

Team	x̄_A	x̄_B	s²_A	s²_B	pooled s²	SE	t	p-value	Decision	vs truth
Team 01	7.42	7.28	1.024	0.691	0.857	0.414	0.34	0.739	do not reject	correct
Team 02	7.29	6.89	1.628	0.739	1.183	0.486	0.82	0.422	do not reject	correct
Team 03	7.02	7.26	0.593	1.060	0.827	0.407	-0.59	0.562	do not reject	correct
Team 04	7.78	7.75	0.853	0.769	0.811	0.403	0.07	0.941	do not reject	correct

Team	$\bar{x}$_A	$\bar{x}$_B	s^2_A	s^2_B	pooled s^2	SE	t	p-value	Decision	vs truth
Team 05	6.89	7.74	1.088	0.416	0.752	0.388	-2.19	0.042	reject H0	**Type I error**
Team 06	7.21	7.16	0.701	1.029	0.865	0.416	0.12	0.906	do not reject	correct
Team 07	7.94	7.36	0.329	0.569	0.449	0.300	1.93	0.069	do not reject	correct
Team 08	6.90	7.31	0.289	1.252	0.770	0.393	-1.04	0.310	do not reject	correct
Team 09	7.27	7.26	1.307	0.798	1.052	0.459	0.02	0.983	do not reject	correct
Team 10	7.23	7.66	1.829	0.898	1.364	0.522	-0.82	0.421	do not reject	correct
Team 11	7.47	7.49	0.309	0.790	0.549	0.331	-0.06	0.953	do not reject	correct
Team 12	7.71	7.82	1.017	0.675	0.846	0.411	-0.27	0.792	do not reject	correct

Step 3: one-way ANOVA (df = 2 and 27)

(a) df: treatments $k - 1 = 2$, error $n - k = 27$, total $n - 1 = 29$. SSE = Total SS − SST. MST = SST / 2, MSE = SSE / 27, $F = MST / MSE$. (b) Yes: with equal sample sizes, $MSE = (s^2_A + s^2_B + s^2_C) / 3$. Small differences are rounding. (c) H_0: $\mu_A = \mu_B = \mu_C$, H_a: at least one mean differs. Reject if $F > 3.35$. Step 4 (b): three tests at $\alpha = 0.05$ each give a larger overall chance of a Type I error; ANOVA keeps it at 0.05 with one test and uses all 30 scores to estimate the variance.

Team	x̄_C	s²_C	grand mean	Total SS	SST	SSE	MST	MSE	F	p-value	Decision	vs truth
Team 01	8.40	1.258	7.700	34.20	7.45	26.75	3.72	0.991	3.76	0.0363	reject H0	correct
Team 02	8.44	0.929	7.540	42.61	12.95	29.66	6.47	1.099	5.89	0.0075	reject H0	correct
Team 03	8.24	1.927	7.507	40.58	8.35	32.22	4.18	1.193	3.50	0.0445	reject H0	correct
Team 04	8.51	0.868	8.013	26.11	3.70	22.41	1.85	0.830	2.23	0.1268	do not reject	**Type II error**
Team 05	7.95	0.854	7.527	27.52	6.30	21.22	3.15	0.786	4.01	0.0299	reject H0	correct
Team 06	8.55	0.867	7.640	35.81	12.43	23.38	6.22	0.866	7.18	0.0032	reject H0	correct
Team 07	8.48	1.760	7.927	30.20	6.27	23.92	3.14	0.886	3.54	0.0431	reject H0	correct
Team 08	8.66	0.880	7.623	38.75	16.96	21.79	8.48	0.807	10.51	0.0004	reject H0	correct
Team 09	8.52	0.551	7.683	34.40	10.50	23.90	5.25	0.885	5.93	0.0073	reject H0	correct
Team 10	8.39	1.279	7.760	42.93	6.88	36.05	3.44	1.335	2.58	0.0947	do not reject	**Type II error**
Team 11	8.51	0.814	7.823	24.29	7.07	17.22	3.54	0.638	5.55	0.0096	reject H0	correct
Team 12	8.63	0.502	8.053	24.79	5.05	19.75	2.52	0.731	3.45	0.0463	reject H0	correct

Note: SSE may differ from Total SS – SST by 0.01 in the table because each number is rounded separately.

Step 4: discussion, and what a good hand-in (e) says (graded, 3 points)

(a) Sampling variation: each team has a different sample of 10, so $\bar{x}$ and s differ; with n = 10 the test has little power.

1. Uses the team's own results: names at least two of the three decisions correctly.
2. Says B is not shown to differ from A (so the new milk is not worth a change), and C scores higher (if the team's ANOVA rejects).
3. Weighs the NT$8 extra cost of C against the higher score, and admits the limit of 10 customers per recipe.

Either "choose C" or "keep A and test C on more customers" earns full marks if the reasons use the data.

After the reveal

(d) Step 1: not rejecting is a Type II error (A really is above 7). Step 2: rejecting is a Type I error (A and B are the same). A small sample has a large standard error, so real differences are often missed (Type II errors are common); the Type I risk stays at α.

Common mistakes to cold-call on

- Using z = 1.645 instead of t = 1.833 in Step 1.
- df = 20 instead of 18 in Step 2.
- Dividing by n = 30 instead of 1/10 + 1/10 in the Step 2 standard error.
- H_a in Step 3 written as "all means differ" instead of "at least one differs".
- "Do not reject" read as "A is not above 7" or "the recipes are the same".